TEAM NEREUS

NEREUS

Nautical Environmental Rover for Ecosystem Restoration and Underwater Sustainability

> "We clean the beaches. But who cleans beneath the water?"

NEREUS is an intelligent underwater environmental rover designed to remove submerged debris and monitor water quality in lakes, rivers, and coastal areas — protecting ecosystems that surface-level cleanup leaves behind. Aligned with **SDG 6**, **SDG 11**, and **SDG 14**.





THE PROBLEM

The Hidden Pollution Crisis

Accumulating Debris

Plastic bottles, cans, and fishing lines settle on riverbeds and lake floors — invisible to surface cleanup efforts.

Harm to Aquatic Life

Submerged waste entangles wildlife, degrades water quality, and breaks down into harmful microplastics.

Dangerous to Clean

Manual removal requires divers — a process that is time-consuming, expensive, and often hazardous.

OUR SOLUTION

Meet NEREUS



NEREUS is a **turtle-inspired semi-submersible environmental rover** designed for smaller water bodies — lakes, rivers, reservoirs, canals, and beach shorelines.

EcoShell Design

Protects electronics while keeping the rover partially above water for stability and communication.

Active Collection

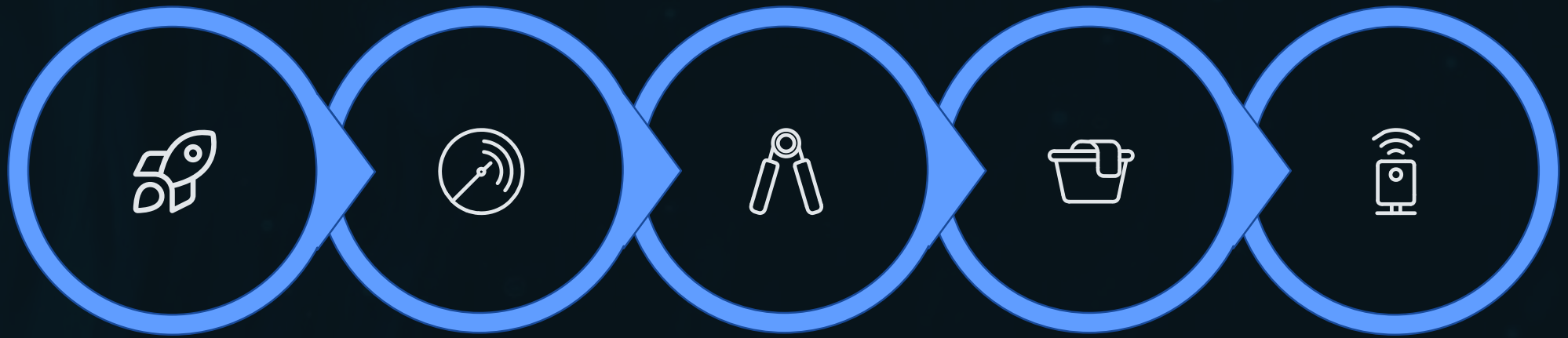
Unlike inspection-only ROVs, NEREUS actively removes submerged debris in real time.

Water Quality Monitoring

Simultaneously tracks pH, turbidity, and temperature during every cleaning mission.

HOW IT WORKS

From Navigation to Collection



Navigate Detect Collect Store Monitor

NEREUS navigates autonomously, detects debris via sonar, collects it with a hybrid gripper, stores it in a rear basket, and continuously monitors water quality — all in a single integrated mission.

HARDWARE & TECHNOLOGY

Engineering the Platform

Core Hardware

- **ESP32** microcontroller — central command unit
- **Dual waterproof thrusters** for propulsion
- **BTS7960** motor driver for high-power control
- **Servo motors** operating the robotic gripper
- **EcoShell** with closed-cell EVA foam for buoyancy and low center of gravity

Sensor Suite

Sonar

Obstacle & debris detection

MPU6050 IMU

Balance & orientation

pH Sensor

Water acidity monitoring

Turbidity

Water clarity measurement

Temperature

Thermal water profiling

Leak Detection

Internal safety monitoring

INNOVATION

Why NEREUS Stands Apart

Turtle-Inspired Design

Biomimetic EcoShell provides natural stability and protects all internal components.

Hybrid Gripper

Collects plastic, metal, and glass — not limited to magnetic materials.

Live Monitoring

Real-time water quality data gathered during every cleaning operation.

Modular & Affordable

Designed for future upgrades: AI vision, autonomous navigation, and IoT integration.



IMPACT

Aligned with Global Goals



NEREUS is purpose-built to advance the UN Sustainable Development Goals by making underwater environmental stewardship **accessible, affordable, and scalable** for communities worldwide.

SDG 14 — Life Below Water

Direct removal of submerged pollution protecting aquatic ecosystems.

SDG 6 — Clean Water & Sanitation

Continuous monitoring helps maintain cleaner freshwater resources.

SDG 11 — Sustainable Cities

Empowers local authorities to maintain cleaner rivers, lakes, and canals.

THE ROAD AHEAD

NEREUS Is Just the Beginning



AI Vision

Automatic debris recognition and classification



Autonomous Routes

Self-guided cleaning missions without operator input



Solar Docking

Solar-powered recharging stations for extended deployment



Swarm Robotics

Coordinated multi-rover cleaning operations



Cloud Monitoring App

Real-time environmental data for authorities and NGOs

> "NEREUS doesn't just clean what we can see — it protects what lies beneath."